

Haitong Ma

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Education

Harvard University	Cambridge, MA
• PhD student in Electrical Engineering	2022.08 – present
▸ Advisor: Prof. Na Li	
Tsinghua University	Beijing, China
• M.S. in Automotive Engineering	2019.08 – 2022.06
▸ Thesis: Safe Control with Statewise Constrained Reinforcement Learning	
▸ Recipient of the National Scholarship (top < 1%)	
• B.S. in Automotive Engineering	2015.08 – 2019.06

Internship Experiences

Google Research	Mountain View, CA
Student Researcher. Advisor: Guy Tennenholtz, Bo Dai	2025.05 – Present
• Developed efficient reinforcement learning (RL) algorithms for discrete diffusion models, achieved SOTA performance on multiple challenging discrete decision tasks such as DNA generation models and long-horizon control in Atari games.	
• Engineered diffusion-based RL on large-scale distributed training to control tokamak for nuclear fusion (really challenging continuous control tasks) in collaboration with Google DeepMind.	
Mitsubishi Electric Research Laboratories	Cambridge, MA
Research Intern. Advisor: Yebin Wang	2024.06 – 2024.08
• Designed hierarchical RL for high-precision real-time motion planning and control of tractor-trailer systems. The RL-based planner achieved millimeter-level accuracy.	
Robotics Institute, Carnegie Mellon University	Pittsburgh, PA
Visiting student. Advisor: Prof. Changliu Liu	2021.06 – 2021.08
• Developed safe reinforcement learning algorithms that can jointly learn safe policies and safety guarantees with theoretical justifications, won L4DC 2022 best paper finalist (only 3 per year).	

Research Interests

Areas:

- **RL / Diffusion:** RL with diffusion policies, Representation-based RL, Constrained/Safe RL
- **Robot Learning / Safe Autonomy:** Imitation Learning, Skill representation, Sim-to-real transfer, Control barrier functions, Reachability
- **Control / Optimization:** Bayesian optimization, Multi-agent coverage, Optimal control

Applications:

- Autonomous driving, Robotic locomotion, Drones, Medical robotics

Honors and Awards

• Harvard SEAS nomination for Two Sigma & Jump Trading PhD Fellowship	2025
• Learning for Dynamics & Control Conference (L4DC) Best Paper Award Finalists (3/176)	2022
• Outstanding Graduate and Master Thesis of Tsinghua University	2022
• Best Student Paper Award, IEEE International Conference on Intelligent Transportation (2/597)	2021

Selected Publications

* means equal contributions. The full list of publications can be found on my [Google scholar](#).

Reinforcement Learning / Diffusion Models

- **H. Ma**, O. Nabati, A. Rosenberg, B. Dai, O. Lang, I. Szpektor, C. Boutilier, N. Li, S. Mannor, L. Shani, G. Tenenholz. "Reinforcement Learning with Discrete Diffusion Policies for Combinatorial Action Spaces." *arXiv*, 2025. [arXiv](#)
- T. Chen, **H. Ma**, K. Wang, N. Li, B. Dai. "One-Step Flow Policy Mirror Descent." *arXiv*, 2025. [arXiv](#)
- **H. Ma**, T. Chen, K. Wang, N. Li, B. Dai, "Efficient Online Reinforcement Learning for Diffusion Policy." *International Conference on Machine Learning (ICML)*, 2025. [paper](#)
- Yang Hu, **H. Ma**, B. Dai, N. Li, "Efficient Duple Perturbation Robustness in Low-rank MDPs." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025. [arXiv](#)
- **H. Ma**, C. Liu, S.E. Li, S. Zheng, W. Sun, J. Chen, "Learn Zero-constraint-violation Safe Policy in Model-Free Constrained Reinforcement Learning." *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2024. [arXiv](#)
- Dongjie Yu*, **H. Ma***, S.E. Li, J. Chen, "Reachability Constrained Reinforcement Learning." *International Conference on Machine Learning (ICML)*, 2022. [arXiv](#)

Robot Learning / Safe Autonomy

- **H. Ma**, B. Dai, Z. Ren, Y. Wang, N. Li, "Offline Imitation Learning upon Arbitrary Demonstrations by Pre-Training Dynamics Representations." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. [arXiv](#) [webpage](#)
- **H. Ma**, Z. Ren, B. Dai, N. Li, "Skill Transfer and Discovery for Sim-to-Real Learning: A Representation-based Viewpoint," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024. [webpage](#) [arXiv](#) [video](#)
 - **Oral presentation (18.4%)**
- H. Zheng, **H. Ma**, S. Zheng, S.E. Li, J. Wang, "Synthesize Efficient Safety Certificates for Learning-Based Safe Control using Magnitude Regularization." *IEEE International Conference on Robotics and Automation (ICRA)*, 2024. [arXiv](#)
- **H. Ma**, Z. Ren, B. Dai, N. Li, "A Hierarchical Approach for Tractor-trailer Motion Planning Using Graph Search and Reinforcement Learning," *European Control Conference (ECC)*, 2024. [paper](#)
- **H. Ma**, C. Liu, S.E. Li, S. Zheng, J. Chen, "Joint Synthesis of Safety Certificate and Safe Control Policy using Constrained Reinforcement Learning." *Learning for Dynamics & Control Conference (L4DC)*, 2022. [arXiv](#)
 - **Oral Presentation, Best paper award finalists (3 out of 176)**

Optimal Control / Bayesian Optimization

- T. Ren*, Z. Ren*, **H. Ma**, B. Dai, N. Li, "Stochastic Nonlinear Control via Finite-Dimensional Spectral Dynamic Embedding." *IEEE Transactions on Automatic Control (TAC)*, 2025. [arXiv](#)
- R. Zhang*, **H. Ma***, N. Li, "Multi-agent Coverage Control with Transient Behavior Consideration," *Learning for Dynamics & Control Conference (L4DC)*, 2024. [arXiv](#) [video](#)
- **H. Ma**, T. Zhang, Y. Wu, F.P. Calmon, N. Li, "Gaussian Max-value Entropy Search for Multi-agent Bayesian Optimization." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023. [arXiv](#) [video](#)

Applications

- T. Zhang*, S. Kim*, J. Charton*, **H. Ma**, K. Kim, N. Li, Q. Li, "Autonomous Robotic Ultrasound System for Liver Follow-up Diagnosis: Pilot Phantom Study." *arXiv*, 2024. 
- **H. Ma**, J. Chen, S.E. Li, Z. Lin, Y. Guan, Y. Ren, S. Zheng, "Model-based Constrained Reinforcement Learning using Generalized Control Barrier Function." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021.  
- Y. Guan*, Y. Ren*, **H. Ma**, S.E. Li, Q. Sun, Y. Dai, B. Cheng, "Learn Collision-free Self-driving Skills at Urban Intersections with Model-Based Reinforcement Learning." *IEEE International Conference on Intelligent Transportation (ITSC)*, 2021.  
 - ▶ **Best student paper award (2 out of 597)**

Academic Services

Served as a reviewer for

- **Peer-reviewed journals:** IEEE Transactions on Automatic Control (TAC), IEEE Transactions on Neural Networks and Learning Systems (TNNLS), Automatica.
- **Peer-reviewed conferences:** ICML, NeurIPS, ICLR, L4DC, CDC, ICRA, IROS, ITSC.

Teaching and Mentoring Experiences

Teaching Fellow, Harvard SEAS

2023.09 – 2023.12

ES155: Systems and Control.